Joint-Embedding Predictive Architectures and Fluid-Dynamic Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

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***Abstract*—Reconstructive anomaly detectors score raw observations, making them sensitive to benign sensor variation. Joint-Embedding Predictive Architectures (JEPAs) forecast latent summaries, but a scalar residual threshold can be distorted by drifting coordinates. We pair a multi-horizon JEPA with an independently calibrated 224-bit circuit-breaker panel. Its 32 coordinates yield four horizon-error magnitudes plus horizon mean, fluctuation, and endpoint difference. Optional post-training quenching modifies selected predictor rows and residual-calibration buffers using nominal data. On a separately trained 19-stream development benchmark, the quenched panel achieves buffered point-wise F1 of 0.262 and causal false-alarm rate of 1.6 percent, with 59.6-step detection delay. In a repeat 19-stream ablation, the unquenched panel achieves causal F1 of 0.071 versus 0.013 for scalar scoring; a 160-dial subset reaches 0.098. These runs are not paired. On five selected streams, horizons of one to 64 steps show panel gains over scalar scoring, including zero structural reporting lag. Across 118 untouched held-out streams, the unquenched three-horizon panel raises causal F1 from 0.083 to 0.143 and event recall from 51 to 73 percent. In a matched 162-stream four-horizon comparison, quenching cuts nominal causal false alarms by 8.2 percent relative while slightly lowering causal F1. Threshold calibration depends on quantile and nominal history; slow ramps and single-sensor faults remain limitations.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Fluid-dynamic decomposition, Boolean circuit-breaker panels, Representation learning.**

I. Introduction

Spacecraft, chemical plants, server farms, and wearable medical devices report health through dozens of continuous sensor channels [1]–[4]. When a valve sticks, a reaction wheel drifts, or a patient's gait freezes, an automated monitor must detect the fault promptly while learning healthy behavior from normal history alone, because labeled faults are rare. Equally important, the monitor must keep false alarms low during nominal operation so operators can trust its alerts.

Why Observation-Space Reconstructors Suffer High False-Alarm Rates: Deep detectors such as TimesNet [5], TranAD [6], and Anomaly Transformer [7] reconstruct the raw sensor signal and alert when reconstruction error is large. Raw telemetry, however, contains unforecastable sensor jitter, thermal drift, and disparate channel scales. Summed across 25 to 55 channels, these misses form a heavy-tailed residual with frequent spikes (Table I). Calibrated on healthy data via Streaming Peaks-Over-Threshold (SPOT) [8] and evaluated on later healthy data, reconstructors raise false alarms during 22%–31% of normal operation; a random alerter firing as often would catch 60.0% of anomaly events, exceeding TimesNet's 58.8%.

The Bottleneck of Scalar Score Aggregation: Joint-Embedding Predictive Architectures (JEPAs) [9], [10] avoid fitting raw noise by compressing each telemetry window into a 32-dimensional unit-sphere summary and predicting future summaries in latent space, cutting false alarms to 0.6%–2.4%. Yet existing latent detectors still sum all coordinate errors into one scalar score compared against one threshold. If one or two of the 32 coordinates shift slightly between training and calibration—variance-dominating coordinates we term *hijacker dials*—they dominate the sum and inflate the threshold, so faults on the remaining 30 coordinates go undetected: causal Point-F1 falls to 0.050–0.052 and average delay exceeds 240 steps.

*Four-Stage Architecture (*Laminar-JEPA*):* We replace scalar score aggregation with a four-stage pipeline, *Laminar-JEPA* (Fig. 1). First, a causal dilated encoder maps context and target windows onto a 32-dimensional spherical latent space via layer-normalization (LayerNorm) [11] and forecasts nested target prefixes; training-residual covariance is filtered by a resolvent precision transform. Second, by structural analogy to Reynolds averaging [12], we decompose 32 coordinate error trajectories into four horizon errors, their mean, fluctuation, and endpoint difference. Third, these $7 \times 32 = 224$ statistics feed independently calibrated circuit-breaker dials and 32 right-edge coincidence relays (*Reynolds 224-Bit* when unquenched). Fourth, optional post-training *weight-space quenching* scales selected predictor rows and adjusts calibration buffers before panel calibration.

A. Contributions

We separate our evaluation into a 19-stream (and 33-stream) development cohort for ablations and a disjoint 118-stream untouched held-out cohort (within a 141- to 162-stream archive across 15 datasets) providing primary generalization evidence (Table II):

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1) *Panel, quenching, and held-out cohorts:* On the reproducible 19-stream ablation cache, the unquenched panel raises causal Point-F1 from 0.013 to 0.071 versus scalar scoring ($p = 0.0023$), while the 160-dial subset reaches 0.098. The separately trained 33-stream benchmark yields 0.262 buffered Point-F1 and 59.6-step causal delay for quenched *Laminar-JEPA*; these are not paired measurements on the rechecked cache. On 118 untouched held-out streams, the unquenched three-horizon panel raises causal Point-F1 from 0.083 to 0.143 and event recall from 51.2% to 72.8%; four-horizon Laminar-JEPA reaches 62.2% event recall. In a matched four-horizon 162-stream comparison, quenching lowers causal held-out false-positive rate (FPR) by 8.2% relative but slightly lowers causal Point-F1.
2) *Mechanism and latency limits:* Matched-backbone controls in Table III(c) measure latent bottleneck and calibration-drift effects; scalar spectral reweighting offers no reliable benefit. The rechecked dial ladder in Panel (a) and a separate residual-mean-buffer recentering study test scoring alternatives without identifying one universally best configuration. An exploratory five-stream sweep over $S \in \{1, 4, 16, 32, 64\}$ finds 1.9–4.3 times the scalar causal Point-F1 at $S = 1, 4$; it cannot establish cross-domain latency generalization.
3) *False-alarm reduction and diagnostic fixed-budget recall:* On the 19 development streams, latent prediction holds autonomous causal FPR at 0.6%–1.7% versus 22.2%–24.3% for reconstructors ($N = 7$ dataset $p = 0.016$). Under a non-deployable post-hoc diagnostic control at $f^* = 1\%$ test FPR, the panel catches 58.4% of anomaly events versus 21.7% for TimesNet and reaches 0.658 buffered precision-recall area under the curve (PR-AUC) on NASA contextual faults (Tables IV–V, Fig. 4).
4) *Calibration sample size, single-channel branch, and null results:* Proposition 1 bounds the horizon-fluctuation concentration ratio $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ (a structural representation property; scalar R_{LT} reweighting is null, $\Delta\mathrm{F1} = -0.005$). Moving-block bootstrap analyses suggest that longer nominal histories stabilize empirical thresholds, but $N_{\mathrm{eff}} = 300$ does not establish reliability for the 99.9th percentile. Interquartile-range (IQR) and Beta-smoothed calibrators yielded test FPR of 0.47%–1.32% in the evaluated short-history settings. Across 17 multivariate streams and six injected single-sensor fault classes, a complementary channel-level branch raises event recall from 61.4%–80.7% to 93.0%–98.2%; these controlled injections do not establish performance on naturally occurring sensor faults, and slow ramps remain undetected.

Roadmap. Section II reviews related work, Section III details the benchmark, Section IV formulates the model and panel, Section V defines the protocol, Sections VI–VII report results, and Section VIII concludes.

II. Related Work

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timestamps where point-level error

TABLE I
Key Terms.

Term	Meaning
Latent residual	Multi-horizon gap $\tilde{\mathbf{E}} \in \mathbb{R}^{4\times 32}$ between predicted and observed 32D window summaries.
224-bit breaker	Boolean panel of 7×32 per-coordinate tripwires and 32 onset relays replacing 1D scalar thresholding.
Reynolds dials	Per-coordinate horizon mean $\lvert\bar{e}_c\rvert$ and fluctuation σ'_c across nested sub-horizons (32 dials each).
Right-edge relay	Coincidence switch $r_c(t)$ requiring terminal sub-block $[48, 64]$ and endpoint change or fluctuation to trip together.
Hijack guard	Nominal-only per-dial multiplier $\gamma_j^{\mathrm{cal}}\gamma_j^{\mathrm{sc}}$ damping variance-dominating coordinates (*hijacker dials*) on calibration data.
Surgical quenching	Post-training calibration pass removing mean predictor-bias drift and tempering top precision directions.
Buffered / Causal	Retrospective prefix-window voting vs. real-time trailing attribution at $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later healthy residuals, measured by Kolmogorov–Smirnov distance D_{KS}.
Concentration ratio	Structural ratio $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ measuring horizon-fluctuation energy in the leading eigenmode.

$\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [4] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [7] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that trigger frequent false alarms under baseline shift [13].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space [14]. Contrastive detectors construct positive and negative pairs: temporal and contextual contrasting [15] and TS2Vec [16] maximize contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [10] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [17] employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.088 and 0.097 stream Point-F1 for NCAD and DCdetector vs. 0.186 for latent prediction across 130 streams, Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [18], [19] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} from a separate target encoder, preventing collapse via exponential moving average (EMA) target updates [20] or variance-covariance penalties such as VICReg [21], Barlow Twins [22], and batch whitening [23].

Reynolds-Inspired Residual Decomposition and Circuit-Breaker Panels vs. Statistical Process Control: Multi-sensor telemetry reflects physical dynamical trajectories [24]. Neural ordinary differential equations (ODEs) [25], Physics-Informed Neural Networks (PINNs) [26], and Hamiltonian or Lagrangian networks [27], [28] enforce differential conservation

laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [29]. Koopman and operator-theoretic models [30]–[34] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we apply a Reynolds-inspired multi-horizon residual decomposition [12], by analogy, to split multi-horizon latent residuals into per-coordinate horizon mean, fluctuation, and endpoint difference. Classical multivariate statistical process control (SPC) charts such as Hotelling's T^2 [35] and Multivariate Exponentially Weighted Moving Average (MEWMA) [36] quadratic-form-pool whitened coordinates into a single Mahalanobis scalar, just as deep detectors collapse residuals into a 1D score calibrated via Pickands–Balkema–de Haan extreme-value theory [37] and Generalized Pareto Distribution (GPD) fitting in SPOT [8], where a single coordinate with out-of-sample variance inflation under serial correlation [38] dominates the scalar threshold. Conversely, banks of univariate Shewhart or Cumulative Sum (CUSUM) charts [39] monitor raw channels or single statistics under Bonferroni or false-discovery-rate (FDR) multiplicity corrections [40], but lack multi-horizon kinematic perspectives, per-dial drift quarantine guards, and right-edge coincidence relays; soft attention ensemble voting across dials adds no benefit over Boolean quorums (Table III(b)). We evaluate all detectors using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm curves [13], [41], [42].

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [13], and mTSBench [45], expanded in archive-wide evaluations to 141 qualifying multivariate streams across 12 datasets and 162 streams across 15 datasets (Table II). Thirty-three streams enter the primary protocol (19 non-GHL development streams and 14 GHL streams); 12 are reserved with explicit accounting below. The suite spans four domains: *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [46] (K=38), where NASA's archive [4] classifies each SMAP/MSL sequence as an out-of-range *point* fault or in-range *contextual* fault (Table V(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [43] at 64 Hz (K=9, 72 episodes, where S=64 equals 1 s). *Industrial and infrastructure telemetry* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry at 1-minute resolution (K=9, 37 anomalies), Metro I-94 hourly traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six CICIDS network streams (K=72). Section VI-B evaluates six single-channel fault classes across all 17 multivariate development streams ($K > 1$).

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$) and labeled test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2}. Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by a 64-step isolation gap (Section V-A). Primary 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$) and $K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$ (Table IV(c)); among the 12 reserved streams, five have $T_{\text{nom}} \in [764, 1{,}492]$ ($N_{\text{cal}} \leq 101$ across CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), six CICIDS streams have K=72, $N_{\text{test}} \geq 1.36 \times 10^5$, and GECCO ($K$=9) enters the archive cohorts. To eliminate subset-selection bias, Table VI evaluates all 141 qualifying multivariate streams across 12 datasets ($N_{\text{cal}} \geq 100$ and ≥ 1 test anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 OPPORTUNITY, GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN) at three seeds, of which 118 streams were never inspected during development, and evaluates four-horizon Laminar-JEPA across 162 streams spanning 15 datasets (adding 14 GHL, Secure Water Treatment (SWaT), Water Distribution (WADI), and reserve streams; 139 untouched streams).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [42]; and (ii) streams within multi-stream archives share hardware platforms. We therefore treat dataset clusters ($N \in \{7, 9, 12, 15\}$) as co-primary statistical units alongside streams, reporting Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and leave-one-dataset-out (LODO) verification (Section V-A).

IV. Methodology: Joint-Embedding Predictive Architectures and 224-Bit Circuit-Breaker Panels

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to binary alarm decisions. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and splits the stream at index t into a historical *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$) and a future *target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to causal decision timestamp $\tau = t + S - 1$). To separate slow drift from high-frequency vibration before cross-channel mixing, multichannel inputs ($K > 1$) pass through a causal depthwise 1D convolutional stem ($K \leq 1$ uses an identity bypass) and are split across $J = 4$ smooth dyadic shells on a 256-point real fast Fourier transform (rFFT) grid normalized into an amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), filtered by per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax router into a causal dilated temporal convolutional network (TCN) [48].

Unlike reconstructive detectors that penalize raw waveform error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$, we map windows into a compact 32-dimensional latent space ($D = 32$) using a Joint-Embedding

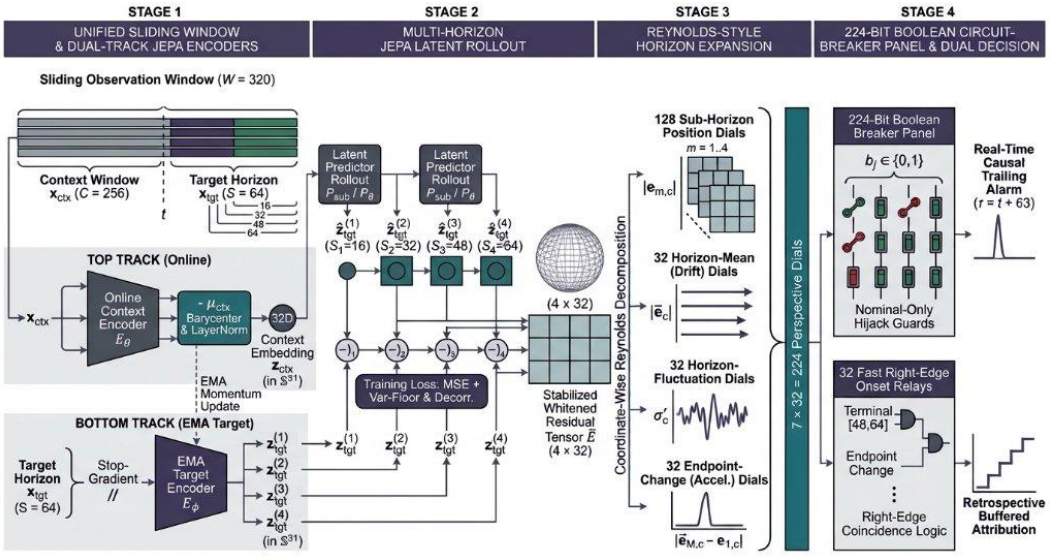


Fig. 1. Overview of the four-stage *Laminar-JEPA* pipeline: **(1)** sliding window (C=256, $S \in \{1..64\}$) and dual encoders (E_θ, E_ϕ) with barycenter centering and LayerNorm; **(2)** nested rollout (P_{sub}, P_θ) under variance-floor and decorrelation loss, whitening, and surgical quenching into $\tilde{\mathbf{E}} \in \mathbb{R}^{4\times32}$; **(3)** Reynolds-inspired expansion into $7 \times 32 = 224$ dials ($|\tilde{e}_{m,c}|$, $|\bar{e}_c|$, σ'_c, $|\tilde{e}_{M,c}-\tilde{e}_{1,c}|$); and **(4)** 224-bit breakers ($b_j \in \{0,1\}$) and 32 onset relays emitting causal (τ=t+S−1) or prefix-buffered alarms.

Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ to approximately radius-$\sqrt{D}$ layer-normalized vectors; the normalization epsilon prevents exact sphere membership. Target weights ϕ track online weights θ via an exponential moving average (EMA), $\phi_k \leftarrow m\phi_{k-1} + (1-m)\theta_k$ with momentum $m = 0.996$ (0.995 in archive cohorts) [18], [20]. Because deep dilated TCN stacks can smooth sharp onset spikes while non-stationary regimes shift the latent mean, Stage 1 introduces two operations inside both encoders:

1) *Multi-depth functional hierarchy tapping:* To preserve fast transient spikes alongside long-range context, a tap head extracts $L_{\text{tap}} = 3$ feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B\times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 48$) from the shallow dyadic stem, mid-depth TCN block, and terminal TCN block. Concatenating the channel-centered terminal state and temporal root-mean-square (RMS) energy of each tap forms $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B\times 6d_{\text{model}}}$, which is projected to $\mathbb{R}^D$ and fused with the deep representation $\mathbf{z}_{\text{deep}}$ via a bounded gate $g = 0.5\,\sigma(\theta_g) \le 0.5$:
$$\mathbf{z}_{\text{fused}} = (1-g)\,\mathbf{z}_{\text{deep}} + g\ \mathrm{LN}\big(W_2\,\phi_{\text{GELU}}(\mathrm{LN}(W_1\mathbf{h}_{\text{tap}}))\big). \tag{1}$$
2) *Synchronized barycenter centering:* An online running centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ (updated at rate 0.05 on training batches) is copied synchronously to the EMA target encoder and subtracted from both context and target vectors prior to terminal LayerNorm, $z = \mathrm{LN}(\mathbf{z}_{\text{fused}} - \bar{\boldsymbol{\mu}}_{\text{ctx}})$. This attenuates learned common-mode offsets without guaranteeing invariance to unseen drift.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Variance-Floor Decorrelation Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a full-window forecast $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$, trained with predictive mean-squared error (MSE) and a log-determinant volume term $-\frac{\beta}{D}\ln\det(\boldsymbol{\Sigma}+\epsilon\mathbf{I})$ ($\beta = 0.05$) related to VICReg [21]. LayerNorm limits each vector's scale but does not bound its temporal tail kurtosis; pooling an $S = 64$ window into one vector can average out brief faults.

To resolve how a trajectory evolves across the target window, Stage 2 of *Laminar-JEPA* uses $M = 4$ nested prefixes $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ with $S_m = \min(S, \max(2, \lfloor mS/M \rfloor))$ for $m < M$ and $S_M = S$ ($\{16, 32, 48, 64\}$ at $S = 64$). Each is encoded by E_ϕ into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. At $S = 1$, all four targets contain the same observation; distinct residuals arise from the predictor rollout, not from distinct target horizons. Starting from $\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$, a recurrent sub-horizon predictor P_{sub} rolls out sequential forecasts $\hat{z}_{\text{tgt}}^{(m)} = \mathrm{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \dots, M-1\}$ using learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the full-horizon anchor P_θ predicts $\hat{z}_{\text{tgt}}^{(M)} = \mathrm{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ ($m < M$) and $w_M = \frac{1}{2}$, training minimizes multi-horizon MSE alongside a non-collapse matrix penalty $\boldsymbol{\Delta}_\gamma$:
$$\mathcal{L}_{\text{MHR}} = \sum_{m=1}^{M} \frac{w_m}{D}\Big(\big\|\hat{z}_{\text{tgt}}^{(m)} - \mathrm{sg}\big(z_{\text{tgt}}^{(m)}\big)\big\|_2^2 + \big\|\boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big)\big\|_F^2\Big). \tag{2}$$
Here $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ penalizes collapse without requiring $B > D$. Its diagonal entries are $\sqrt{\max(0, \gamma - \sigma_d)}$, where σ_d uses the batch-biased variance and 10^{-4} stabilization ($\gamma = 0.75$). Off-diagonal entries are the sample covariances $\Sigma_{dj}/\sqrt{2}$ ($d \neq j$), computed with the $B-1$ divisor. The distinct variance conventions matter for exact correspondence with the objective. At $B = 16 < D = 32$, the measured effective rank $\mathrm{RankMe}(z_{\text{tgt}})$ rises from 11.6 to 19.8 [49]. Archive experiments use three-horizon *Speculative-JEPA* on 141 streams (the fourth prefix is interpolated, 192 statistics) and four-horizon *Laminar-JEPA* on 162 streams.

C. Stage 3: Calibration-Stabilized Whitening, Surgical Weight-Space Quenching, and 224-Dial Expansion

Why does collapsing a multi-horizon residual tensor into a 1D scalar score degrade real-time detection? Stacking the $M =$

4 prediction errors $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ forms $\mathbf{E}(t) \in \mathbb{R}^{M\times D}$ across $MD = 128$ coordinates. On training windows, the sample residual covariance has eigendecomposition $\hat{\mathbf{\Sigma}}_{\text{res}} = \mathbf{U}\,\mathrm{diag}(\lambda_k)\mathbf{U}^\top$. The deployed resolvent precision is

$$\mathbf{\Omega} = \mathbf{U}\,\mathrm{diag}\left(\frac{\lambda_k}{(\lambda_k + 10^{-3})^2}\right)\mathbf{U}^\top.$$

This $MD \times MD$ precision matrix is positive semidefinite. Applying $\mathbf{\Omega}^{1/2}$ to residuals centered by the training mean $\bar{\boldsymbol{\mu}}_{\text{res}}$ gives $\mathbf{E}^{\text{white}}(t) \in \mathbb{R}^{M\times D}$. This attenuates near-null modes rather than imposing unit training variance in every coordinate. Nominal calibration shifts can still inflate selected coordinates and dominate a scalar norm's tail. Stage 3 addresses this through two mechanisms:

1) *Weight-space quenching (*Laminar-JEPA*) and calibration-stabilized whitening:* A nominal calibration shift can inflate a few residual coordinates. The implemented post-training intervention does not invert a LayerNorm Jacobian or apply an additive tangent-space bias correction. It ranks coordinates using training-to-calibration residual shifts, variance inflation, within-calibration drift, autocorrelation, and nominal trip frequency. For each of the top $k \in \{4, 2\}$ coordinates, it scales the selected output row and bias of both predictor multilayer perceptrons and their output LayerNorm gains by $a_c = \mathrm{clip}((1 + 0.5\max(0, r_c))^{-1}, 0.35, 0.85)$, where r_c is the standardized coordinate instability score. It also moves the corresponding learned horizon embeddings by 0.35 times the calibration-minus-training mean residual shift and blends each selected residual-mean buffer entry 75% toward its recalculated calibration mean. For the $MD \times MD$ precision buffer, cross-coordinate entries incident to selected coordinates retain 15% of their value; an eigendecomposition projects the result onto the positive-semidefinite cone before per-coordinate calibration-RMS rescaling. These are direct parameter and buffer edits, not an analytic cancellation of predictor bias. The candidate is accepted only if forward and reversed nominal half-split checks lower internal false alarms or near-threshold excess without increasing forward false alarms, while both main quorums retain at least 92% of their original values; otherwise the original model is restored. This nominal-only check does not guarantee preservation of fault responses (Section VI-A). Any coordinate with $\mathrm{RMS}_{\text{cal}}^{\text{white}}(m, c) > 1.10$ is then blended toward its diagonal-standardized residual $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/(\mathrm{RMS}_{\text{cal}}^{\text{raw}}(m, c) + 10^{-4})$:

$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t),$$
$$w_{m,c} = \frac{1}{1 + 2\max\big(0, \mathrm{RMS}_{\text{cal}}^{\text{white}}(m, c) - 1.10\big)^2}. \quad (3)$$

2) *Reynolds-inspired multi-horizon expansion into $7D = 224$ dials:* By structural analogy to Reynolds decomposition in fluid turbulence [12] (applied along the discrete look-ahead axis $m \in \{1, \ldots, M\}$ of each coordinate $c \in \{1, \ldots, 32\}$), we decompose $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into horizon mean $\bar{e}_c(t) = \frac{1}{M}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)$ and zero-mean fluctuation $e'_{m,c}(t) = \tilde{e}_{m,c}(t) - \bar{e}_c(t)$, forming $7D = 224$ non-negative dials $\mathbf{d}(t) \in \mathbb{R}_{\geq 0}^{224}$: (a) *sub-horizon position magnitudes (4D = 128)* $d_{\text{pos},m,c}(t) = |\tilde{e}_{m,c}(t)|$ $(m \in \{1..4\})$; (b) *horizon-mean drift (D = 32)* $d_{\text{drift},c}(t) = |\bar{e}_c(t)|$; (c) *horizon fluctuation (D = 32)* $d_{\text{fluc},c}(t) = \sigma'_c(t) = \sqrt{\frac{1}{M}\sum_{m=1}^{M} e'^2_{m,c}(t)}$; and (d) *endpoint difference (D = 32)* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$.

Proposition 1 describes the dials after stabilization: subtracting their horizon mean separates residual energy into a mean and a zero-mean part. It also bounds endpoint difference and spectral concentration for that same residual matrix, not for the raw or independently transformed horizons; these structural bounds do not guarantee detection accuracy.

Proposition 1 (Structural Characterization of Multi-Horizon Residual Energy and Spectral Concentration)**.** *Let $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ $(M \geq 2)$ denote the stabilized multi-horizon residual field with horizon mean $\bar{\mathbf{e}} = M^{-1}\tilde{\mathbf{E}}^\top \mathbf{1}_M \in \mathbb{R}^D$ and zero-mean fluctuation matrix $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M \bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. Then: (i)* Coordinate-wise energy orthogonality and endpoint bound: *For every coordinate $c \in \{1, \ldots, D\}$, $\frac{1}{M}\sum_{m=1}^{M} d^2_{pos,m,c}(t) = d^2_{drift,c}(t) + d^2_{fluc,c}(t)$, and $d_{accel,c}(t) \leq \sqrt{2M}\, d_{fluc,c}(t)$. (ii)* Spectral concentration bounds for the horizon Gram matrix: *Let $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ have ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ ($\mathbf{E}'^\top \mathbf{1}_M = \mathbf{0}$). For any $\gamma > -1/2$, the spectral concentration ratio (following Lieb–Thirring power-sum moments [50]),*

$$R_{LT}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^{M} \lambda_m^{\gamma+1/2}} \in \Big[\frac{1}{M-1}, 1\Big] \quad (\mathbf{E}' \neq \mathbf{0}), \quad (4)$$

is scale-invariant, attaining $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) iff fluctuation energy is isotropic across all $M - 1$ modes, and 1 iff $\mathrm{rank}(\mathbf{E}') = 1$. Empirically, scalar reweighting by R_{LT} yields a null effect (Section VI).

Proof. Expanding $\frac{1}{M}\sum_{m=1}^{M}(\bar{e}_c + e'_{m,c})^2$ with $\sum_{m=1}^{M} e'_{m,c} = 0$ yields the energy identity, and Cauchy–Schwarz on $e'_{M,c} - e'_{1,c}$ gives $d^2_{\text{accel},c} \leq 2(e'^2_{M,c} + e'^2_{1,c}) \leq 2M\, d^2_{\text{fluc},c}$, proving (i). For (ii), $\mathbf{G}\mathbf{1}_M = \mathbf{0}$ forces $\lambda_1 = 0$, so $\lambda_M^{\gamma+1/2} \leq \sum_{m=2}^{M} \lambda_m^{\gamma+1/2} \leq (M-1)\lambda_M^{\gamma+1/2}$, yielding (4). □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

Rather than summing the 224 dials into a scalar norm, Stage 4 (Fig. 1) equips each dial with an independent binary circuit breaker calibrated on nominal data. Because all 224 dials derive from the same 4×32 residual field, their trips are correlated by construction (median effective rank 65 of 224 on the development cohort, with joint trip-count variance $7.2\times$ that of independent trials), so alarm quorums are calibrated directly on the empirical distribution of simultaneous nominal trip counts:

1) *Per-dial standardization and nominal hijack guards:* Each dial $j \in \{1, \ldots, 224\}$ is standardized by its nominal calibration median m_j and scaled median absolute deviation (MAD) $s_j = 1.4826\,\mathrm{MAD}(d_{j,\text{cal}}) + 10^{-4}$ into $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine any individual coordinate that drifts during calibration, each dial's 98.5th-percentile tripwire τ_j is scaled by a two-factor nominal hijack guard $\gamma_j = \gamma_j^{\text{cal}}\,\gamma_j^{\text{sc}}$:

$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\,\gamma_j^{\text{cal}}\,\gamma_j^{\text{sc}}, \quad (5)$$

where γ_j^{cal} is the ratio of second-half to first-half calibration 98th percentiles whenever > 1.35 (1 otherwise), and γ_j^{sc} is $(m_j + s_j)/(1.5\, s_{\text{group},j})$ whenever > 1.25 (1 otherwise, relative to perspective-group median $s_{\text{group},j}$).

2) *32 fast right-edge onset relays:* To trigger rapidly when an abrupt fault enters the right edge of the S-step window while earlier prefixes ($m \in \{1,2\}$) still observe nominal data, each coordinate $c \in \{1,\dots,32\}$ defines a strict edge switch $b_{p,c}^{\text{edge}}(t) = \mathbb{I}\big(z_{p,c}(t) > q_{0.993}(z_{p,c,\text{cal}})\,\gamma_{p,c}^{1.25}\big)$ and fires its onset relay $r_c(t) \in \{0,1\}$ when a terminal-prefix trip ($m = 4$) coincides with acceleration, fluctuation, or prefix $m = 3$:

$$r_c(t) = b_{\text{pos},4,c}^{\text{edge}}(t) \wedge \big(b_{\text{accel},c}^{\text{edge}}(t) \vee b_{\text{fluc},c}^{\text{edge}}(t) \vee b_{\text{pos},3,c}^{\text{edge}}(t)\big). \quad (6)$$

3) *Causal trailing vs. buffered attribution and Boolean quorums:* Under *causal trailing attribution*, switches on window $[t, t+S)$ are emitted at its right endpoint $\tau = t+S-1$ ($\mathcal{F}_\tau$-measurable, with structural lag $S-1$ and empirical delay $\Delta t = t_{\text{alm}} - t^*$). Under *retrospective buffered attribution*, the 224-dial vector routes position votes to $[t, t+S_m)$, drift and fluctuation votes to $[t, t+S)$, and endpoint-difference votes to $[t+S_3, t+S)$; the separate 128-dial counter and onset relays instead average uniformly over the whole S-step window. At $S = 1$, the buffered endpoint-difference span is empty. Summing active breakers gives $N_{224}(\tau)$, $N_{128}(\tau)$, and $N_{\text{edge}}(\tau)$ (ranking score $N_{\text{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\text{edge}}(\tau)$ for PR-AUC). Binary alarms trigger when any counter exceeds its nominal-calibration quorum:

$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \;\vee\; N_{128}(\tau) > K_{128} \;\vee\; N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \quad (7)$$

where $K_{\text{edge}} = \max\big(1.25\, q_{0.999}(N_{\text{edge,cal}}), 2.5\big)$, *224-Bit Breaker* sets $(K_{224}, K_{128}) = \big(\max(1.12\, q_{0.995}, 5.0), \max(1.12\, q_{0.995}, 3.5)\big)$ on nominal trip counts, and *224-Bit Strict* tightens them to $\big(\max(1.18\, q_{0.9985}, 5.5), \max(1.20\, q_{0.9985}, 4.0)\big)$.

1D Scalar Collapse Controls: To isolate the gain of the 224-bit panel over 1D scalar scoring on identical backbones (Section VI), we compare against a soft coordinate-saliency pooler $g_d(\mathbf{v}) = (1-\alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}$ with softmax weights $w_i \propto \exp(|v_i|/(\tau_g \sigma_{\mathbf{v}}))$ ($\sigma_{\mathbf{v}} = \|\mathbf{v}\|_2/\sqrt{d} + 10^{-6}$, $\alpha = \tau_g = 0.5$). Single-horizon TS-JEPA scores $\mathcal{S}_{\text{TS}}(t) = g_D(\mathbf{e}(t))$, and 1D Multi-Horizon Reynolds JEPA combines full-field and peak-prefix norms via $\mathcal{S}_{\text{env}}(t) = \sqrt{\frac{1}{2}(g_{MD}(\tilde{\mathbf{e}}(t))^2 + \max_m g_D(\tilde{\mathbf{e}}_m(t))^2)}$ (scaled by $1 + \alpha_{\text{LT}} \max(0, \tanh(0.5\, z_{\text{LT}}(t)))$ for $\alpha_{\text{LT}} \in \{0.12, 0\}$), calibrated via extreme-value SPOT [8], [37], [51] ($u = p_{98}, q = 10^{-3}$).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup, Two-Stage Nominal Calibration, and Statistical Design

All architectures operate on a streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), trained on nominal windows at stride 16 and evaluated at stride 1. Context and target windows are encoded into $D = 32$ spherical representations by the multiscale front-end and causal dilated TCN, fused with the three-depth tap head (1), and centered by the running context barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$ prior to terminal LayerNorm. Models are trained using AdamW [52] (learning rate 10^{-3}, weight decay 10^{-4}) with zero per-dataset tuning: batch 16 for 15–20 epochs on the 19- and 33-stream cohorts, and batch 16–64 for 10–12 epochs with gradient clipping at 1.0 and cosine annealing on the archive cohorts.

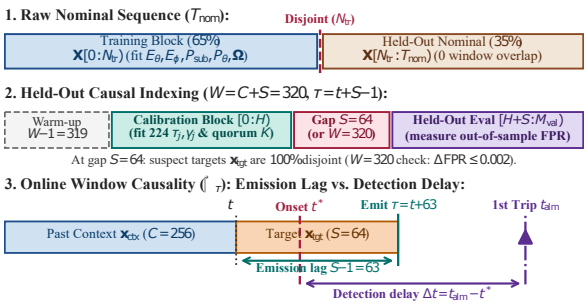


Fig. 2. Two-stage held-out evaluation protocol (C=256, S=64, W=320): (Row 1) chronological train/validation split, (Row 2) boundary-isolated nominal calibration and evaluation blocks fitting the 224 dial tripwires τ_j, hijack guards γ_j, and quorums ($K_{224}, K_{128}, K_{\text{edge}}$), and (Row 3) structural emission lag ($S-1$) versus empirical fault-detection delay (Δt).

a) Two-Stage Nominal Calibration Protocol and Boundary Isolation: Each stream's nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$): (1) *Stage A (Training fit on $\mathbf{X}_{tr}$):* Standardization parameters, encoder/predictor weights, running barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and resolvent precision matrix $\boldsymbol{\Omega} \in \mathcal{S}_+^{MD}$ are fit on $\mathbf{X}_{\text{tr}}$; archive checkpoint selection uses unsupervised predictive loss on the nominal calibration portion. The later nominal evaluation block is not used for this selection. (2) *Stage B (Circuit-breaker calibration and held-out verification on $\mathbf{X}_{val}$):* Sliding windows of length $W = 320$ inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2) yield $M_{\text{val}} = N_{\text{val}} - W + 1$ valid window timestamps. Splitting at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $[0 : H)$ and *held-out evaluation block* $[H + S : M_{\text{val}})$, separated by an $S = 64$-step isolation gap $[H : H + S)$ so calibration and evaluation share zero target observations (a full $G = 320$ gap check changes causal FPR by < 0.25 percentage points). On $[0 : H)$, the panel fits $w_{m,c}$ (3), per-dial medians m_j and MAD scales s_j, hijack guards γ_j, tripwires τ_j (5), and quorums ($K_{224}, K_{128}, K_{\text{edge}}$) (7) (or τ_{SPOT} for 1D baselines), which are frozen and evaluated on $[H + S : M_{\text{val}})$ and on the test stream. Throughout the paper we distinguish nominal calibration window count $N_{\text{cal}} = H$ from autocorrelation-adjusted *effective independent sample size* $N_{\text{eff}} = N_{\text{cal}}/\tau_{\text{int}}$ ($\tau_{\text{int}} = 1 + 2\sum_{k \geq 1} \rho_k \approx 3.23$).

b) Hierarchical Statistical Units, Dataset-Clustered Inference, and Hypothesis Testing: Because streams within multi-stream archives share hardware and operating environments, we report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Stream-level and dataset-clustered inference:* Seeds are averaged within stream before two-sided paired Wilcoxon tests and 95%

TABLE II
CONSOLIDATED EXPERIMENTAL COHORTS AND VALIDATION ROLES.

Cohort	Str.	Sets	Primary Purpose	Role
Ablation 19	19	6	Ladder, quenching, 15-arm study	Dev.
Primary 33	33	7	Core (19 dev. + 14 GHL)	Dev.+GHL
Tuning Set 23	23	7	Panel & quenching selection	Dev.
Archive 141	141	12	3-horizon (23 dev. + 118 held)	Mixed
Untouched 118	**118**	**9**	**Primary generalization check**	**Held-Out**
Non-Exath. 101	**101**	**8**	**Cluster-balanced held-out**	**Held-Out**
Laminar 162	162	15	4-horizon quenching (139 held)	Mixed
Ext. Screen 130	130	11	NCAD, DCdetector, Anom. Tr.	Mixed

stream-bootstrap confidence intervals (CI; 2,000 resamples). For multiplicity reporting, the two focal held-out causal Point-F1 contrasts (panel vs. Speculative-Alien-JEPA and vs. TS-JEPA) form one family; this family was not preregistered. Benjamini–Hochberg correction [40] gives adjusted stream-level $p = 3.1 \times 10^{-13}$ and 6.4×10^{-11}, respectively; both dataset sign-flip tests give adjusted $p = 0.016$ across nine families. Quoted p-values elsewhere are raw and exploratory. Dataset-cluster bootstrap CIs and non-Exathlon held-out subsets ($n = 101$) assess sensitivity to shared dataset origin. (2) *Retrospective 12-fold LODO calibration-selection audit:* Selecting the calibrator on $D-1$ datasets across the 141-stream archive to maximize causal Point-F1 subject to FPR $\leq 5\%$ selects the 224-bit rule in all $12/12$ folds. This analysis reuses the archive and is not an additional independent validation of the 118-stream comparison.

B. Consolidated Cohort Organization, Development vs. Held-Out Separation, and Baseline Fairness

Table II consolidates the experimental cohorts and separates development streams from independent held-out validation streams. Because the calibration constants in (5)–(7) and surgical quenching settings were selected during development on 23 streams (the 19 non-GHL streams plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, and `OPPORTUNITY S1-ADL3`), the 19- and 23-stream cohorts are explicitly designated as *development* studies, whereas the 118 **untouched streams** (101 non-Exathlon streams across 8 datasets, and 139 untouched streams in the 162-stream Laminar evaluation) were never inspected during design and serve as the *primary independent evidence of generalization*.

Standardized Baseline Fairness Protocol: Scalar baselines (TS-JEPA, TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], and DCdetector [17]) use the common $65\%/17.5\%/17.5\%$ split, training-only standardization, isolation gaps, and SPOT calibration ($u = p_{98}$, $q = 10^{-3}$); the panel instead calibrates Boolean quorums on the same nominal block. Common SPOT does not guarantee equal suitability for every baseline. At $K = 25$, parameter counts span 206–649k; the panel adds no trainable parameters. Native published windows on five streams alter TimesNet and TranAD outcomes, so the common-window and native-window comparisons answer different questions. Runs used PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.88 ms/window, 188 MB); code: https://github.com/quaid281/NCAD.

C. Multi-Tier Operational Evaluation Metrics

To guard against point-adjustment inflation and threshold gaming [13], [41], we report: (i) *primary accuracy and false-alarm outcomes* comprising held-out nominal FPR (FPR_{hold}), unadjusted Point-F1 (with Precision and Recall in Section VII), and threshold-free PR-AUC (from $N_{\text{tot}}(\tau)$); (ii) *deployable vs. diagnostic operating characteristics* comparing autonomous circuit-breaker operating points (*224-Bit Strict* and *224-Bit Breaker*), deployable calibration quantiles (q^*, Fig. 4(b)), and, strictly as a non-deployable post-hoc diagnostic control, matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table III(c)); and (iii) *secondary operational endpoints* comprising Dataset-Macro Point-F1, event recall (ER), mean detection delay $\Delta t = t_{\text{alm}} - t^*$, and the constant-alarm reference $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at prevalence π.

VI. ABLATION STUDIES AND ANALYSIS

A. Ablation 1: Perspective Expansion Ladder, Circuit-Breaker Components, and Weight-Space Quenching

Table III(a),(b) and Fig. 3(a) trace the expansion and component choices on the currently reproducible 19-stream cache. The separately trained study in Panel (c) is not paired to this cache:

1) *From scalar collapse to sub-horizon switches:* The full panel raises causal F1 from 0.013 to 0.071 on this cache ($p = 0.0023$), but its buffered difference of $+0.033$ has a stream-bootstrap interval spanning zero ($p = 0.53$). The 160-dial position-plus-drift rule scores higher causal F1 (0.098) than all 224 dials. Scalar R_{LT} modulation remains null in the separate fixed-backbone study.
2) *Component ablations:* Removing relays changes mean causal delay from 90.1 to 90.7 steps; removing prefix routing changes buffered delay from 83.7 to 115.4 steps. Without hijack guards, causal F1 rises from 0.071 to 0.073 and delay falls to 70.1 steps on this cache, so these ablations do not establish a universal benefit for the guards, relays, or additional dial groups. None removes the $S-1$ causal reporting lag.
3) *Quenching components and sensitivity ($k_{rogue} \in \{0, \ldots, 12\}$):* On the development streams, the separate residual-mean-buffer recentering ablation (without predictor-weight edits) raises causal event recall from 56.5% to 64.3% and buffered F1 from 0.224 to 0.256, at the cost of higher nominal FPR. Precision tempering alone lowers buffered F1 to 0.135, while RMS stabilization (3) raises causal F1 to 0.099. Sweeping $k_{\text{rogue}} \in \{1, 2, 4, 8, 12\}$ shows that the selected $k \in \{4, 2\}$ nominal-split-gated intervention lowers archive-wide causal FPR across 162 streams from 6.71% to 6.17% ($p = 0.0011$), while more aggressive edits ($k \in \{8, 12\}$) reduce causal PR-AUC from 0.292 to 0.223. This ablation does not isolate an additive predictor-bias correction.

Statistical detector controls on the same cached runs: A position-plus-drift subset reaches causal F1 0.098 at nominal held-out FPR 0.006, versus 0.071 and 0.008 for the full panel; its paired F1 interval spans zero. A top-k vote reaches F1 0.078 at FPR 0.010. Hotelling T^2, MEWMA, and CUSUM score F1 0.271–0.282 but incur held-out FPR 0.304–0.594. These development controls do not establish a matched-alert-rate advantage over statistical process control.

TABLE III
MULTI-PERSPECTIVE CIRCUIT-BREAKER EXPANSION AND COMPONENT ABLATIONS (A,B: RECOMPUTED 19-STREAM CACHE); SEPARATELY TRAINED MATCHED-BACKBONE STUDY (C).

(a) Perspective Expansion Ladder ($1 \rightarrow 224$ **Dials,** n=19)

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
Perspective Expansion Stage	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
1D Scalar Collapse (1 Dial)	.013	.191	**.315**	.413	.459	**.550**	166.2	143.1
32-Bit Full Horizon (m=4)	.050	.135	.255	.381	.500	.243	123.4	65.5
128-Bit Sub-Horizons (m=1..4)	.063	.216	.294	**.449**	.556	.462	110.6	**38.8**
160-Bit (+32 Reynolds Drift)	**.098**	.198	.304	.429	**.591**	**.550**	**74.8**	108.3
192-Bit (+32 Reynolds Fluc.)	.067	.192	.299	.443	.573	.453	79.1	101.6
224-Bit (+32 Endpoint/Relays)	.071*	**.224**	.299	.448	.564	.506	90.1	83.7

* Versus 1D, causal F1 difference $+0.057$ (stream-bootstrap 95% CI $[0.016, 0.120]$; Wilcoxon $p = 0.0023$, 12/2 wins/losses). Buffered difference $+0.033$ (CI $[-0.017, 0.093]$, $p = 0.53$). The scalar spectral-modulation ablation in the separately trained study remains null.

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
Component / Quorum Variant	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
224-Bit Breaker (Full Panel)	.071	.224	.299	.448	**.565**	**.506**	90.1	83.7
224-Bit Strict (High Quorum K_{strict})	.047	.168	.299	.448	.494	.439	207.6	**54.3**
w/o 32 Right-Edge Relays	.068	.225	.299	**.452**	**.565**	**.506**	90.7	83.7
w/o Prefix Routing (Unif. Smear)	.071	.222	.299	.440	**.565**	.497	90.1	115.4
w/o Hijack Guards (γ_j=1)	**.073**	**.226**	**.299**	.448	**.565**	**.506**	**70.1**	83.3

Panels (a,b): 19 streams × 3 seeds on the same whitened-residual cache, without the stabilized blend in (3). Panel (c) uses a separate training run; cross-panel comparisons are not paired.

(c) Matched-Backbone Architectural Comparison (n=19 **Development Streams**)

	Tail / Drift		Held-Out FPR		Point-F1		PR-AUC		Causal Event Recall			
Architecture Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	Auto.	q^*=0.05	f^*=0.01	f^*=0.05
Reynolds 224-Bit Breaker	2.30	0.179	0.017	0.132	0.138	0.228	**0.312**	**0.411**	**0.607**	0.774	**0.584**	**0.728**
Reynolds 224-Bit Strict	2.30	0.179	0.011	0.098	0.116	0.190	**0.312**	**0.411**	0.534	0.774	**0.584**	**0.728**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Reynolds (1D Scalar)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Reynolds (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

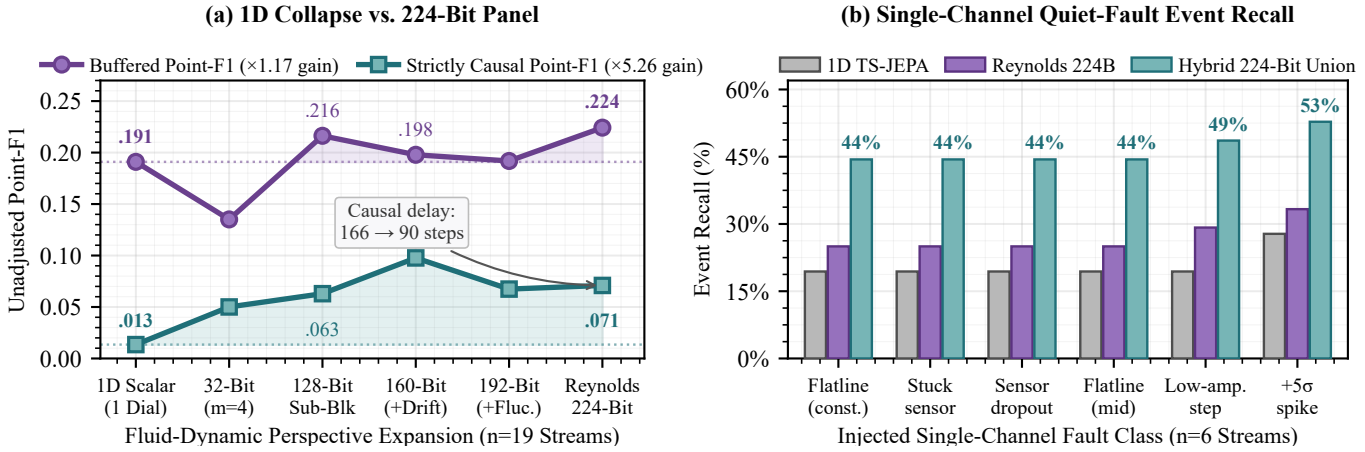


Fig. 3. Development studies. **(a)** Causal and buffered Point-F1 across the six-step expansion on cached whitened residuals from 19 streams (three seeds; no stabilization blend). **(b)** Event recall for injected single-channel faults in a separate six-stream study, comparing 1D TS-JEPA, 224-Bit Breaker, and their channel-level union.

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

When a single sensor out of K freezes or drifts while the remaining $K-1$ channels fluctuate normally, the isolated sensor perturbs the pooled pre-LayerNorm vector by only $O(1/K)$ before spherical normalization rescales the embedding onto $\sqrt{D}\mathbb{S}^{D-1}$. On six injected streams (Fig. 3(b)), 1D TS-JEPA catches 19.4% of quiet-fault events, standalone *Reynolds 224-Bit* reaches 25.0%–29.2%, and a union with a rolling 32-step variance-collapse monitor reaches 44.4%–52.8%. To evaluate a broader channel-level remedy across all 17 multivariate development streams ($K \in [5, 55]$, excluding univariate Yahoo), we pair the 224-bit panel with a complementary channel-level branch that monitors three causal per-channel statistics calibrated at $q_{0.999}$ on $\mathcal{D}_{\text{cal}}$ with drift guards γ_c^{cal}: (i) rolling 32-step variance collapse $\max(0, 1-s_c(t)/\text{med}_{\text{cal}}(s_c))$, (ii) causal first-order autoregressive (AR(1)) prediction residuals $|x_{t,c} - \hat{\phi}_{1,c}x_{t-1,c} - \hat{\phi}_{0,c}|$, and (iii) rolling 32-step mean bias $|\bar{x}_{t,c}^{(32)} - \mu_{\text{cal},c}|$. Across six single-channel fault classes injected into the test windows of all 17 streams, adding the variance-collapse monitor lifts event recall from 61.4% to 93.0% on variance collapse, 63.2% to 93.0% on sensor freezing, and 73.7% to 96.5% on sensor saturation (tripling to quintupling causal F1 to 0.376–0.414). Adding the full AR(1) and bias branch further lifts event recall on single-channel prediction shocks (80.7% → 98.2%, causal F1 0.139 → 0.327), gradual sensor degradation (68.4% → 93.0%, F1 0.080 → 0.154), and isolated sensor bias (70.2% → 93.0%, F1 0.095 → 0.195). On unmodified test streams, however, monitoring $3K$ channel statistics in parallel raises nominal causal FPR from 1.5%

(standalone 224-Bit) to 6.8% (variance-collapse union) and 19.4% (full channel branch); thus, the variance-collapse union provides a practical safeguard for freeze and saturation faults, while full channel-residual monitoring requires cross-channel quorum gating when K is large.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Retraining Multi-Horizon Reynolds JEPA with and without 4-shell rFFT splitting across five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, and `GECCO`, three seeds) shows a clear dimensionality threshold: frequency-shell splitting substantially improves detection on high-dimensional spacecraft arrays ($K = 55$ `MSL M-1` buffered F1 $0.085 \rightarrow 0.410$; $K = 25$ `SMAP P-3` causal F1 $0.212 \rightarrow 0.321$), where separating slow thermal drift from high-frequency actuator chatter prevents cross-band masking, but lowers accuracy on low-channel streams ($K = 9$ `Daphnet` and `GECCO`), paying off only when $K \geq 25$.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 15 model arms under matched budgets and parameter counts in Table III(c) isolates five mechanisms governing false-alarm stability: (1) *Forecasting vs. reconstructing raw observations:* Using the identical causal dilated TCN encoder to forecast or reconstruct raw sensor waveforms leaves heavy residual tails ($\gamma_2 \approx 6.2$), severe calibration drift ($D_{\text{KS}} \approx 0.45$), and high causal false-alarm rates ($\approx 25\%$) virtually unchanged. (2) *LayerNorm bottleneck in observation space:* Inserting a $D = 32$ LayerNorm bottleneck into the observation forecaster compresses tail kurtosis six-fold ($\gamma_2 = 0.91$) and halves causal false alarms, but predicting raw waveforms still leaves calibration drift high ($D_{\text{KS}} = 0.376$). (3) *Learned latent target projection and whitening:* Moving the prediction target into the learned $D = 32$ spherical space halves calibration drift ($D_{\text{KS}} = 0.214$ in TS-JEPA and 0.179 in Multi-Horizon Reynolds JEPA, $p \leq 0.001$). (4) *Nonlinear TCN vs. linear PCA and Isolation Forest:* Linear principal-component forecasting maintains low false alarms but lacks the nonlinear capacity to capture fault dynamics, while Isolation Forest misses most anomaly events. (5) *Prediction vs. autoencoding, frozen targets, and ℓ_2 scoring:* Latent context autoencoding sacrifices causal accuracy relative to multi-horizon prediction, whereas freezing the target encoder or replacing Mahalanobis whitening with ℓ_2 scoring still keeps causal false alarms below 2%, confirming that false-alarm suppression is an intrinsic geometric property of the normalized latent representation.

VII. Results and Discussion

A. Development 33-Stream and Untouched Held-Out 118/162-Stream Benchmark Results

Following the cohort hierarchy in Table II, Table IV reports the separately trained 33-stream development benchmark under retrospective buffered attribution (Panel (a)), causal streaming (Panel (b)), and per-dataset stratification (Panel (c)). Table VI gives the 118-stream design-held-out comparison and the 162-stream matched quenching study. On a separate 130-stream baseline screen, TS-JEPA reaches stream-mean Point-F1 0.186 versus TranAD 0.152, TimesNet 0.151, DCdetector 0.097, NCAD 0.088, and Anomaly Transformer 0.082. The observed reduction in nominal false alarms with latent scoring is specific to these protocols; spherical normalization alone does not prove a distributional guarantee.

Operational trade-offs: The separately trained development comparison in Table III(c) favors *Laminar-JEPA* for onset delay and buffered Point-F1, whereas the current-cache expansion in Panel (a) favors 160 over 224 dials for causal F1. The unquenched three-horizon panel leads event recall and Point-F1 in the 118-stream held-out comparison; these arms have different backbones. In the matched four-horizon 162-stream archive, quenching lowers mean causal held-out FPR from 6.71% to 6.17% while causal Point-F1 falls from 0.0869 to 0.0839. Selection depends on the desired false-alarm and detection trade-off.

Scope boundaries: Observation-space forecasters and reconstructors have high nominal false-alarm rates in the development comparison. Learned latent targets reduce false alarms under the tested protocol, but this does not show that LayerNorm guarantees light tails on arbitrary data. Both JEPAs and linear autoregressors struggle on the 14 GHL slow-ramp streams: low false-alarm rate without event recall is not detection. Deployment choices must account for sampling cadence, the causal lag $S - 1$, stream-specific nominal drift, and the increased false alarms of the optional single-channel fault branch. No universal minimum calibration history follows from these experiments.

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Comparing Panels (a) and (b) of Table IV exposes why standard 1D latent detectors break down when moved from retrospective buffered attribution to real-time causal streaming. Buffered attribution averages residuals across all 64 overlapping horizon predictions covering each timestamp, smoothing out transient noise and yielding a moderate right tail ($\gamma_2 = 1.25$). Under causal streaming, each decision relies on a single right-edge step without temporal averaging, nearly doubling tail kurtosis ($\gamma_2 = 2.30$). When 128 whitened coordinates are summed into a 1D scalar score, one or two high-variance coordinates push the global SPOT threshold high onto that heavy tail, leaving scalar Reynolds JEPA blind to all but brief anomaly spikes. Replacing the scalar sum with 224 independently calibrated circuit-breaker switches and right-edge onset relays (6) prevents any single drifting coordinate from inflating the thresholds of the remaining 223 dials, sustaining detection across the fault window. Evaluating the analytical event recall of a rate-matched Bernoulli random alerter, $\text{ER}_{\text{rand}}(p) = E^{-1} \sum_{e=1}^{E} [1-(1-p)^{L_e}]$, further clarifies the operational stakes: at our panel's $< 2\%$ causal false-alarm rate, random guessing catches only 28.1% of events, whereas at the 24.3% false-alarm rate of TimesNet, a random coin flip catches 60.0% of anomaly windows by chance alone, surpassing TimesNet's own causal recall.

TABLE IV
SEPARATELY TRAINED 33-STREAM DEVELOPMENT BENCHMARK ACROSS ATTRIBUTION REGIMES (19 NON-GHL, 14 GHL); NOT PAIRED WITH THE CURRENT-CACHE ABLATION IN TABLE III(A,B). BEST IN **BOLD**, SECOND UNDERLINED.

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-perspective fluid-dynamic circuit-breaker architectures												
Laminar-JEPA (224-Bit)	0.107	**0.262**	0.406	**0.664**	89.9	0.016	0.100	**0.625**	**59.6**	**0.002**	0.000	**0.139**
Reynolds 224-Bit Breaker	0.132	0.228	**0.411**	0.631	71.8	0.017	0.138	0.607	105.9	**0.002**	0.000	0.122
Reynolds 224-Bit Strict	0.098	0.190	**0.411**	0.545	97.3	0.011	0.116	0.534	248.0	**0.002**	0.000	0.098
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Reynolds (1D scalar)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Reynolds (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	62.2	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	$F1_{all}$	**224-Bit**	**TS-JEPA**	**Rey. 1D**	**TimesNet**	**TranAD**	**224-Bit**	**TS-JEPA**	**Rey. 1D**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.223	0.145	0.143	**0.229**	0.144	**.007/.047**	.030/.091	.008/.072	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.001/.002**	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	**0.063**	0.000	0.012	0.000	0.000	.010/.038	.004/.017	.003/.098	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.199**	0.161	0.081	0.158	0.151	.014/.131	.054/**.095**	**.004**/.098	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.354	**0.379**	0.267	0.289	0.167	.021/.214	.014/.135	**.006/.093**	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	**0.001**	**0.001**	0.000	.002/.007	.008/.014	.002/.007	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.017	**0.119**	0.008	0.000	0.000	**.000/.000**	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**

Note: In (c), n = streams, K = channels, E = events, π = anomaly rate, $F1_{all} = 2\pi/(1+\pi)$. 95% stream-bootstrap CIs (n=19): 224-Bit Breaker buffered FPR [.068, .198], F1 [.142, .321] and causal FPR [.008, .027], F1 [.068, .224] (vs. Reynolds 1D causal F1 [.013, .097] and TimesNet causal FPR [.123, .375]).

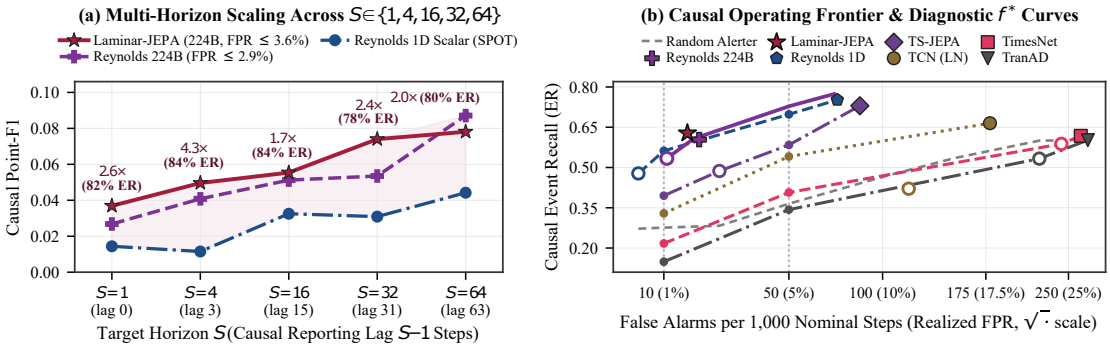


Fig. 4. Multi-horizon scaling ($S \in \{1, 4, 16, 32, 64\}$, causal reporting lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps, n=5 streams) and causal operating frontier (n=19 streams). **(a)** Causal Point-F1 and event recall (ER) across S comparing *Laminar-JEPA (224B)*, *Reynolds 224B*, and 1D scalar SPOT (held-out FPR $\leq 3.6\%$ across all S, vs. $2.5\% \rightarrow 21.3\%$ for TimesNet). **(b)** Causal ER vs. realized false alarms ($\sqrt{\cdot}$ scale) across diagnostic budgets $f^* \in \{1\%, 5\%\}$ (dotted guides), autonomous points (*Laminar-JEPA* star, *Reynolds 224B* cross, *224-Bit Strict* and 1D points open circles), and quantiles $q^* \leq 0.05$.

C. Multi-Horizon 224-Bit Evaluation Across $S \in \{1, 4, 16, 32, 64\}$ and Latency Trade-Offs

Because a target window of length S emits its causal decision at $\tau = t + S - 1$, $S = 64$ imposes a 63-step reporting lag. We retrain on five selected streams at $S \in \{1, 4, 16, 32, 64\}$ to examine the latency trade-off. At $S = 1$, all four predictor states are compared with the same one-step target, so this setting has no distinct target-horizon decomposition; at $S = 4$, the shortest encoded prefix is two steps. In the buffered $S = 1$ setting, the endpoint-difference votes have no routed span. These short-horizon results therefore test the implemented detector, not four distinct horizon views.

In Fig. 4(a), both panel variants exceed 1D scalar SPOT across the five measured horizons; the nominal false-alarm rate stays below 3.6%. At $S = 1$ and $S = 4$, the measured causal Point-F1 is 1.9–4.3 times the scalar score, with event recall above 75%; at $S = 4$, detection delay is 36.7–44.1 steps. At $S = 32$, *Laminar-JEPA* reaches buffered F1 of 0.290 versus 0.182 for 1D, while the unquenched panel peaks in causal F1 at $S = 64$. Five selected streams cannot establish a deployment rule across sampling rates or datasets.

D. Operating Budgets and Contextual vs. Point Anomalies

Fig. 4(b) and Table V examine why multi-perspective latent circuit breaking shifts the detection frontier upward across operating budgets and fault taxonomies:

1) *Dominance across autonomous and diagnostic operating budgets (Fig. 4(b), Table III(c)):* Along the causal recall-versus-false-alarm frontier in Fig. 4(b), the autonomous operating points of *Laminar-JEPA*, *Reynolds 224-Bit*, and *224-Bit Strict* occupy the upper-left low-false-alarm regime without test-label tuning. When all architectures are evaluated at matched post-hoc test false-alarm budgets ($f^* \in \{1\%, 5\%\}$, dotted vertical guides in Fig. 4(b)) as a diagnostic check of tail sensitivity at equal alert rates, the 224-bit panel captures nearly three times as many anomaly events as TimesNet and nearly four times as many as TranAD at a 1% false-alarm budget.

TABLE V
CONTEXTUAL VS. POINT ANOMALY DETECTION (EVENT RECALL (ER)).

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**Auto. ER**	**FPR**	f^*=.01	f^*=.05	f^*=.01	f^*=.05
Reynolds 224-Bit	**0.658**	**0.750**	0.018	**0.820**	**0.920**	0.683	0.944
Reynolds (1D)	0.652	**0.750**	**0.006**	**0.820**	**0.920**	0.683	0.867
Reynolds (α_{LT}=0)	0.641	0.625	**0.006**	**0.820**	**0.920**	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=0.05	f^*=0.01	f^*=0.05	q^*=0.05	f^*=0.01	f^*=0.05
Reynolds 224-Bit	**0.654**	**0.552**	**0.618**	**0.796**	0.688	**0.792**
Reynolds (1D)	0.628	0.498	0.590	0.782	0.683	0.781
Reynolds (α_{LT}=0)	0.640	0.540	0.603	0.785	0.671	0.785
TS-JEPA	0.614	0.520	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Panel (a) stratifies the 10 development SMAP/MSL streams by NASA's ground-truth taxonomy [4] (contextual: SMAP E-1, A-2, A-7, MSL M-1, and C-1 [2100, 2210]; point: SMAP A-1, P-3, D-3, MSL D-14, D-15, and C-1 [550, 750]). Panel (b) stratifies all n=19 non-GHL development streams into in-range contextual ($\max \lvert z\rvert \leq 5\sigma$) vs. out-of-range point ($\max \lvert z\rvert > 5\sigma$) causal events ($f^*$ columns are post-hoc diagnostic controls).

2) *Relational sensitivity to in-range contextual faults (Table V):* Stratifying benchmark events into out-of-range point spikes versus in-range contextual faults under both NASA's ground-truth spacecraft taxonomy [4] (Table V(a)) and a 5σ amplitude boundary across all 19 development streams (Table V(b)) reveals the physical source of this gap. Point anomalies that drive raw sensors beyond 5σ are readily flagged by simple amplitude thresholds and observation-space reconstructors alike. Contextual anomalies, however, occur well inside normal sensor amplitude bounds when cross-channel phase relationships, frequency spectra, or state-transition velocities decouple. Observation-space reconstructors suffer a sharp collapse on these in-range faults, whereas the 224-bit panel monitors horizon drift, fluctuation energy, and velocity shifts across whitened latent coordinates, doubling reconstructor recall on contextual faults while maintaining top performance on point anomalies.

E. Untouched Held-Out Generalization, Calibration Sample-Size Reliability, and Drift Decomposition

Table VI and our archive-wide diagnostic audits establish three foundational results governing out-of-sample generalization and calibration reliability:

1) *Independent generalization across 118 untouched streams and 162 archive streams (Table VI):* On the 118 held-out streams that were never inspected during architectural or calibration design, replacing 1D scalar scoring with the unquenched multi-horizon circuit-breaker panel nearly doubles causal Point-F1 ($p = 1.6\times10^{-13}$, positive across 8 of 9 dataset families with cluster-level $p = 0.008$), lifts event recall by more than twenty percentage points, and cuts causal onset delay by roughly thirty steps across all six multi-stream held-out datasets (Table VI(b)). On these streams the unquenched panel also exceeds Laminar-JEPA in causal Point-F1 (0.143 vs. 0.091), buffered Point-F1 (0.168 vs. 0.149), and causal event recall (0.728 vs. 0.622). Across the full 162-stream, 15-dataset archive, weight-space quenching in four-horizon *Laminar-JEPA* reduces held-out causal false-alarm rates ($p = 0.0011$, dataset-clustered $p = 0.027$) and test false-alarm rates ($p < 10^{-4}$), achieving a median held-out causal FPR of 1.03% (70.4% of streams below 2.0%). The two focal held-out causal Point-F1 contrasts give stream-mean gains $+0.060$ versus Spec.-Alien-JEPA and $+0.071$ versus TS-JEPA; corresponding 95% dataset-bootstrap intervals are $[0.045, 0.069]$ and $[0.048, 0.089]$, respectively. Raw dataset sign-flip p-values are 0.008 and 0.016 (both adjusted $p = 0.016$ for this two-contrast family). In the matched archive comparison, mean event recall decreases from 0.574 to 0.558 on the 161 streams with defined event recall; nominal-only quench selection cannot guarantee anomaly preservation.

TABLE VI
ARCHIVE-WIDE CIRCUIT-BREAKER BENCHMARK (3 SEEDS): 118 HELD-OUT STREAMS NEVER USED IN DEVELOPMENT, FULL 141-STREAM AGGREGATE, AND PER-DATASET BREAKDOWN.

	Pt-F1 ↑		PR-AUC ↑		ER ↑		FPR_{hold} ↓	
(a) Held-Out n=118	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
TS-JEPA (1D)	.072	.142	.154	.202	.484	.458	**.051**	**.104**
Speculative-JEPA (1D)	.076	.148	.172	.245	.509	.577	.054	.194
Spec.-Alien-JEPA (1D)	.083	.146	.174	.242	.512	.573	.059	.199
Spec.-224Bit-Strict	.134	.166	**.183**	**.247**	.662	.673	.073	.181
Spec.-224Bit-Breaker	**.143**†	**.168**‡	**.183**	**.247**	**.728**	**.691**	.083	.210
Laminar-JEPA (224-Bit)	.091	.149	–	–	.622	.609	.073§	.205
All n=141 (incl. 23 dev.)								
Spec.-Alien-JEPA (1D)	.077	.153	.184	.253	.507	.567	.051	.180
Spec.-224Bit-Breaker	.139	.180	.197	.261	.727	.685	.072	.196

		Causal Pt-F1 ↑			Buffered Pt-F1 ↑			Causal FPR_{hold} ↓		
(b) Held-out	n	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**
SMAP	40	**.194**	.129	.091	.201	.193	**.214**	.088	.086	**.065**
SMD	18	**.090**	.028	.042	**.125**	.065	.072	.019	**.006**	.009
Exathlon	17	**.163**	.133	.092	.181	**.186**	.134	.235	.137	**.119**
Daphnet	15	**.125**	.053	.075	**.153**	.125	.105	.085	**.063**	**.063**
MSL	13	**.133**	.051	.074	**.239**	.176	.191	.023	**.007**	.011
OPPORT.	12	**.075**	.018	.024	.066	**.067**	.030	.027	**.003**	.008

†vs. 1D Spec.-Alien: Δ=+0.060 (95% CI [+0.042, +0.079]), 95/15 wins, $p=1.6\times10^{-13}$, positive in 8 of 9 datasets (sign-flip p=0.008); vs. TS-JEPA 85/28, $p=6.4\times10^{-11}$. ‡vs. 1D Spec.-Alien: Δ=+0.021 ([+0.010, +0.034]), $p=1.7\times10^{-4}$, dataset-level p=0.27. §4-horizon Laminar-JEPA vs. unquenched 4-horizon panel: causal FPR_{hold} −8.2% (p=0.0011, dataset p=0.027; median 1.03% on n=162, 1.13% on n=118), FPR_{test} −4.8% ($p<10^{-4}$, dataset p=0.016; median 1.59% on n=162). In (b), 224B = Spec.-224Bit-Breaker, 1D-Al. = 1D Spec.-Alien, TS = TS-JEPA (115 of 118 streams; three single-stream datasets in text).

2) *Calibration sample size and small-sample remedies:* Overlapping windows reduce effective sample size; $\tau_{\text{int}} \approx 3.23$ is a cohort summary, not a constant for each stream or statistic. On the 118 held-out streams, histories below 250 windows have 19.2% mean causal FPR, versus 1.6% for at least 2,500 windows, but the stream populations and drift differ. At $N_{\text{eff}} = 300$, expected effective tail counts are 4.5 at $q_{0.985}$ (dial tripwires), 1.5 at $q_{0.995}$ (main quorums), 0.45 at $q_{0.9985}$ (Strict), and 0.3 at $q_{0.999}$ (edge quorum). None gives a universal threshold-precision or FPR guarantee. Preliminary per-quantile bootstrap cells cover only two streams and cannot establish a population-level reliability boundary; the short-history IQR and Beta-smoothed results are likewise exploratory.
3) *Regime drift vs. tail misfit via three-stage permutation decomposition:* Refitting thresholds across chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and point-wise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) validation splits proves that excess false alarms are not caused by parametric tail misfit within stationary windows, but by inter-block regime drift ($\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$), which drives 71.3% of excess causal FPR in unconstrained TS-JEPA and 78.3%–81.8% in observation-space reconstructors. In our architecture,

running-barycenter centering absorbs slow manifold translation to cut the inter-block drift share four-fold to 18.7%, while per-dial hijack guards γ_j quarantine individual drifting coordinates before they can trigger a global alarm.

VIII. Conclusion and Future Work

We tested latent circuit-breaker panels on 33 development and 118 untouched held-out streams:

1) *Circuit-breaker panel, quenching, and operational trade-offs:* Across the current 19-stream cache, the unquenched panel raises causal Point-F1 over 1D collapse (0.013 to 0.071, $p = 0.0023$), while a 160-dial subset reaches 0.098; its buffered F1 gain is not reliable. On 118 held-out streams, the three-horizon panel improves causal Point-F1 ($p = 1.6\times10^{-13}$, dataset-level $p = 0.008$) but differs from the four-horizon Laminar backbone. In a matched four-horizon comparison across the 162-stream archive, quenching reduces held-out causal false alarms ($p = 0.0011$) while slightly reducing causal Point-F1.
2) *Mechanisms and horizon scaling:* Matched-backbone ablations measure LayerNorm tail compression and barycenter drift suppression; scalar spectral reweighting has no reliable benefit. Across five selected streams, the $S \in \{1, 4, 16, 32, 64\}$ horizon sweep yields a $1.9\times$–$4.3\times$ causal Point-F1 advantage over scalar scoring, including at zero structural reporting lag ($S = 1$). This small sweep does not establish a general latency benefit across datasets.
3) *Applicability boundaries, calibration, and single-channel branch:* The evaluated multichannel streams and nominal histories show a dependence of calibration stability on quantile and sample size. At $N_{\text{eff}} = 300$, the expected effective tail count is 1.5 at $q_{0.995}$ but only 0.3 at $q_{0.999}$; no universal empirical edge-quorum reliability threshold follows. IQR extrapolation and Beta-smoothed quantiles lowered false alarms in the evaluated short-history settings. A complementary channel-level variance and AR(1) residual branch raises recall above 93% on injected single-sensor faults, at the cost of higher nominal alert rates. Naturally occurring sensor failures and slow process ramps (GHL, Metro) remain unresolved.

References

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.

[4] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[8] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[10] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[11] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.

[12] O. Reynolds, "On the dynamical theory of incompressible viscous fluids and the determination of the criterion," *Philosophical Transactions of the Royal Society of London A*, vol. 186, pp. 123–164, 1895.

[13] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[14] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.

[15] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.

[16] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[17] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[18] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[19] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[20] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[21] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[22] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[23] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[24] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[25] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[26] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[27] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[28] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[29] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[30] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[31] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[32] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[33] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[34] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[35] H. Hotelling, "Multivariate quality control, illustrated by the air testing of sample bombsights," in *Techniques of Statistical Analysis*. McGraw-Hill, 1947, pp. 111–184.

[36] C. A. Lowry, W. H. Woodall, C. W. Champ, and S. E. Rigdon, "A multivariate exponentially weighted moving average control chart," *Technometrics*, vol. 34, no. 1, pp. 46–53, 1992.

[37] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.

[38] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[39] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.

[40] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.
[41] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.
[42] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.
[43] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.
[44] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.
[45] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.
[46] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.
[47] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.
[48] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.
[49] Q. Garrido, R. Balestriero, L. Najman, and Y. LeCun, "RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank," in *International Conference on Machine Learning (ICML)*, 2023, pp. 10 929–10 974.
[50] E. H. Lieb and W. E. Thirring, "Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities," in *Studies in Mathematical Physics: Essays in Honor of Valentine Bargmann*. Princeton University Press, 1976, pp. 269–303.
[51] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.
[52] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.